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Tutorial 4: Implement the Radix-2 DIT and DIF FFT algorithm for a given input sequence.

1. Compute FFT of $x[n] = \{1,1,0,0\}$ using decimation in time (DIT) FFT algorithm and decimation in frequency (DIF) FFT algorithm.
2. Derive decimation in time FFT algorithm for 8-point DFT and explain how butterfly structure is used in FFT.
3. Find the 8-point DFT of sequence
 $x(n) = \{1,2,3,4,4,3,2,1\}$ using DIT radix - 2 FFT algorithm
4. Compare DFT and FFT on the basis of computational complexity for $N = 64, 256$
5. Find the 8-point DFT of sequence $x(n) = \{1,2,3,4,4,3,2,1\}$ using decimation in frequency radix-2 FFT algorithm
6. How many computations are required to compute 16pt DFT using DFT and FFT algorithm.
7. Draw signal flow graph of radix-2 DIT FFT algorithm for $N=4$
8. Draw signal flow graph of radix-2 DIF FFT algorithm for $N=4$
9. Why FFT is preferred over DFT?
10. What is the computational complexity of FFT?
11. Compare DIT-FFT and DIF-FFT based on:
 1. Input sequence ,2. Output sequence .3. Butterfly structure,
 4. Twiddle factor multiplication, 5. Computational complexity
12. Discuss practical applications of FFT in communication, image processing, biomedical signal analysis, and audio processing.
